

SAMSUNG

Product Experience Research Sharing Session #8

Competitive Automatic Speech Recognition

Google Voice Action

Samsung Voice Command/ Voice Talk

Apple Siri

Date: 12.06. 2011

Design & Technology Research

Strategic Marketing Research

Samsung Telecommunications America

Galaxy Nexus Google Voice Action & Apple iPhone 4s Siri Product Experience

Fast Facts

Overall Preference & Scores	Google Voice Action	Apple Siri	Remarks
Most Preferred (N=15)	0	15	The results were overwhelmingly negative for Google Voice Action.
System Usability Score	57.67	86.33	SUS Scores over 80 is generally considered as excellent.
Text-to-Speech Performance	81.91%	95.73%	Accuracy of under 90% (stricter criterion under 98%) is unacceptable in automatic speech recognition.
Likelihood of Buying (1-5)	2.60	4.27	1: most unlikely to 5:most likely

Overall Preference & Scores	Galaxy SII Voice Talk & Voice Command	Apple Siri	Remarks
Most Preferred (N=18)	3	15	Among the three respondents who preferred Voice Talk and Voice Command, one did not like simplicity of Apple products, the second one spoke with an accent that Voice Talk & Voice Command could handle well, the third indicated that he would buy Apple iPhone 4s instead because he disliked the form factor of Galaxy SII.
System Usability Score	60.29	85.15	SUS Scores over 80 is generally considered as excellent.
Text-to-Speech Performance	83.73%	94%	Accuracy of under 90% (stricter criterion under 98%) is unacceptable in automatic speech recognition.
Likelihood of Buying (1-5)	2.94	4.17	1: most unlikely to 5:most likely

Galaxy SII Voice talk and Voice command & Apple iPhone 4s Siri Product Experience

Fast Facts

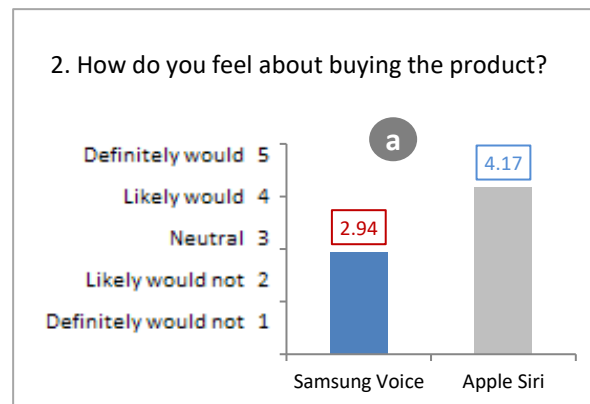
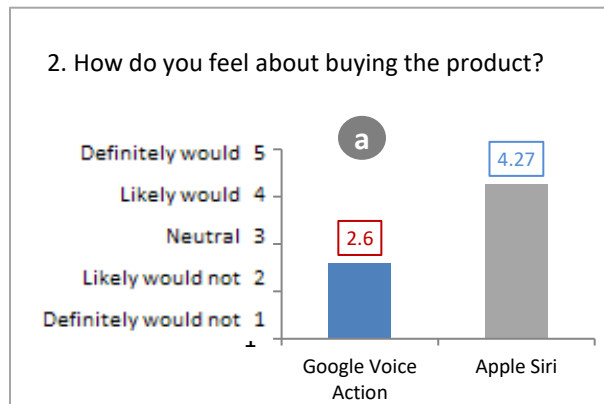
1. Consumers (N=18) selected Apple Siri as most-preferred speech recognition system significantly more than Samsung Galaxy SII Voice Command and Voice Talk after completion of six typical speech-enabled tasks (followed by moderator's tutorial and a 5-minute-plus intensive usage of the speech recognition features and applications of each device) in realistic usage situation.
2. The overall preference, performance and usability scores of Apple Siri was significantly higher than Galaxy SII Voice Talk & Voice Command. Apple Siri also scored either significantly better or trended higher (if there were no statistical differences in a minority of all the measurements) than Samsung Galaxy SII Voice Talk and Voice Command.
3. Consumers (N=15) selected Apple Siri (iPhone 4s) as most-preferred speech recognition system significantly more than Google Voice Action (Galaxy Nexus) after completion of six typical speech-enabled tasks (followed by moderator's tutorial and a 5-minute-plus intensive usage of the speech recognition features and applications of each device) in realistic usage situation.
4. The overall preference, performance and usability scores of Apple Siri were significantly higher than Google Voice Action. Apple Siri also scored either significantly better or trended higher (if there were no statistical differences in a minority of all the measurements) than Google Voice Action.
5. The conversation with Siri also captures a deeper level of emotional engagement with the user than Voice Command/ Voice Talk.
6. Apple Siri poses a major and substantial strategic threat to Samsung and Android automatic speech recognition capability. The threat goes beyond automatic speech recognition and extends into the deeper contextual understanding of Siri, the use of natural language in Siri instead of inflexible command-based syntax of Samsung voice talk or voice commands and the rudimentary artificial/natural intelligence and personality that Siri exhibits.

Product Concepts Evaluation from Consumer Behavioral Research

Google Voice Action, Samsung Voice command/Voice talk vs. Apple iPhone 4s Siri

Insight:

Consumers have substantially higher purchase interest in Apple Siri after using both Voice Action and Siri. They also like Siri better and are more likely to seek more information. The results are very similar for Siri vs. Samsung Voice command/Voice talk.



Automatic Speech Recognition – System Usability Scores

Samsung Voice talk / Voice command & Apple Siri

Insight:

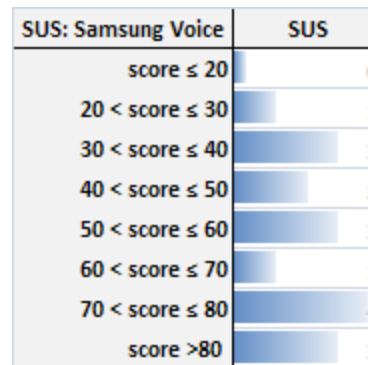
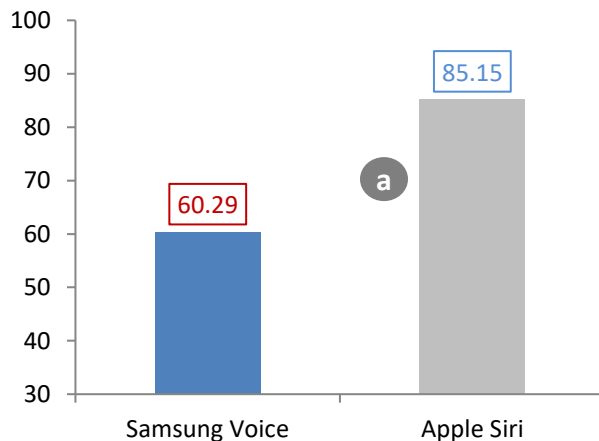
The user experience of Samsung Voice Command/Talk substantially lags behind Apple Siri due to a variety of reasons. Among the most important factors are the rigid and limited vocabulary, lack of a single-step invocation independent of contexts and the lack of contextual & intelligent understanding of speech input.

a

The system usability scores of Apple Siri were significantly higher than Samsung Voice Command and Voice Talk.

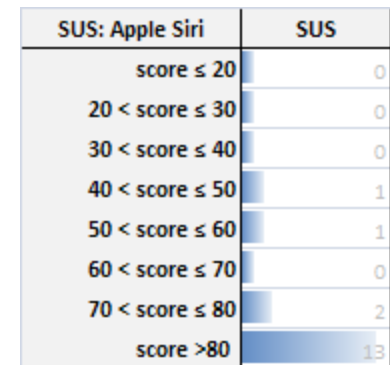
b

The wide distribution of SUS scores is reflective of products with usability issues.



Distribution of number of respondents (total N=18)

b



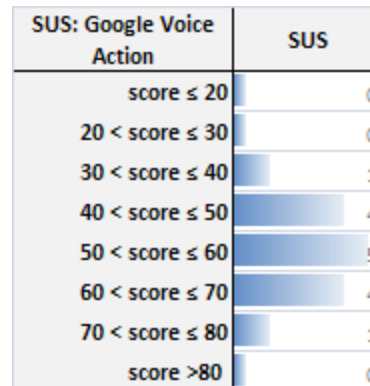
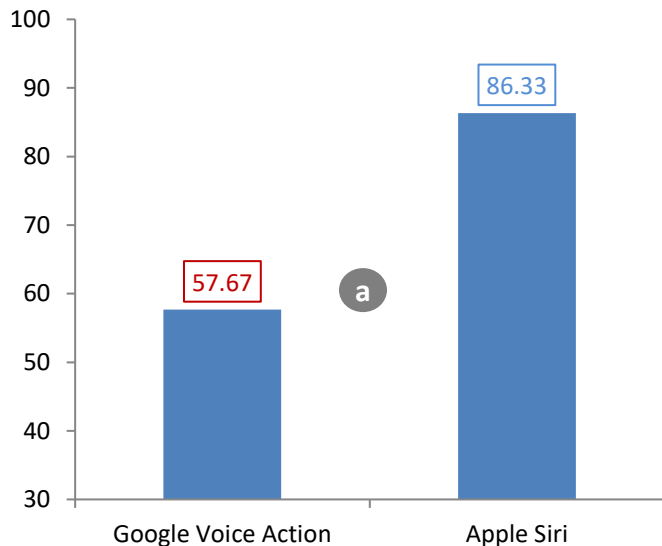
Automatic Speech Recognition – System Usability Scores

Samsung Voice talk / Voice command & Apple Siri

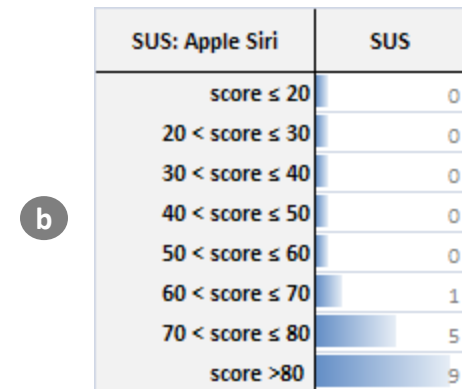
Insight:
The user experience of Google Voice Action substantially lags behind Apple Siri due to a variety of reasons. Among the most important factors are the rigid and limited vocabulary, lack of a single-step invocation independent of contexts and the lack of contextual & intelligent understanding of speech input.

a The system usability scores of Apple Siri were significantly higher than Google Voice Action.

b The wide distribution of SUS scores is reflective of products with usability issues.



Distribution of number of respondents (total N=15)

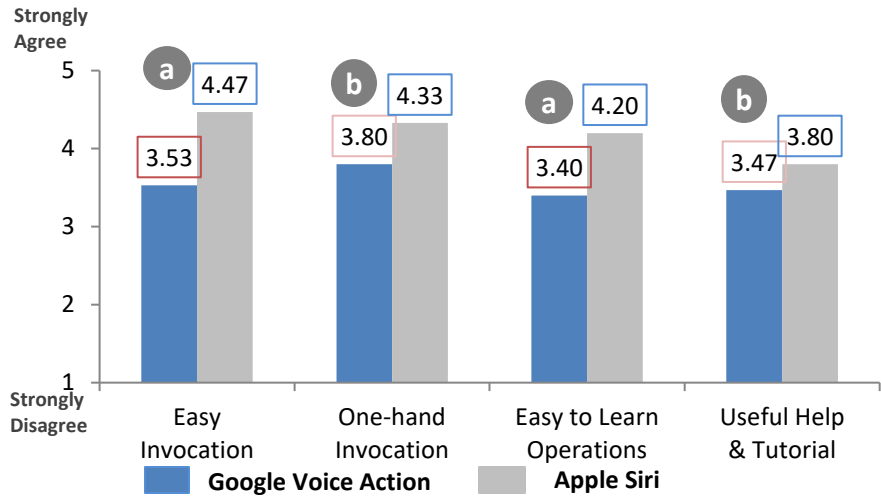


Initial Usage & Impression: Ease of Invocation and Learning

Initial Impressions Ratings with 5-minute exploration of voice-enabled features

Insight:

Siri invocation follows best design practices. It allows direct and effortless invocation of the speech recognizer. Any design that involves more mental and/or physical effort is likely to defeat the purpose of speech recognition.



a Siri scored significantly higher than Voice Action in the ease of invocation and ease of learning.

b Siri scored marginally higher than Voice Action in one-hand invocation and useful help and tutorial.

Initial Product Experience Google Voice Action	Easy Invocation	One-hand Invocation	Easy to Learn Operations	Useful Help & Tutorial
Strongly Agree	5	2	3	1
4	8	9	6	9
3	2	1	6	5
2	2	1	2	0
Strongly Disagree	1	1	0	1

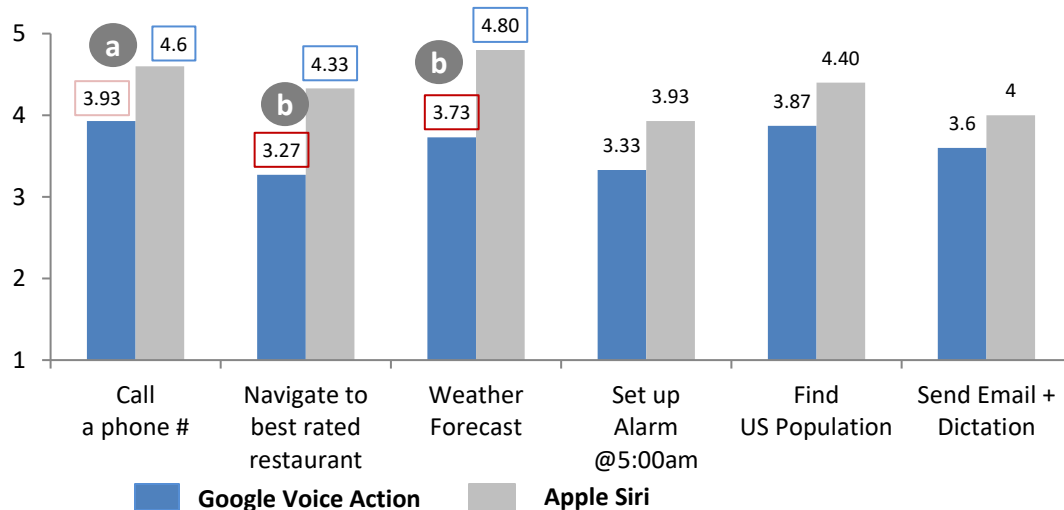
Initial Product Experience Apple Siri	Easy Invocation	One-hand Invocation	Easy to Learn Operations	Useful Help & Tutorial
Strongly Agree	5	8	6	5
4	6	8	8	9
3	1	1	2	3
2	0	0	0	1
Strongly Disagree	1	0	0	0

Distribution of number of respondents (total N=15)

Task Completion - Overview

Tasks Completion Rate

Insight:
Consumers can complete more voice-enabled tasks (with intended results) with Siri than Google Voice Action in less trials . This can be attributed to the recognition accuracy and contextual understanding of Siri (as compared to the “Did you mean” approach in Voice Action).



- a The completion rates for calls from Siri was marginally better than Voice Action.
- b The completion rates for Navigate to best rated Japanese restaurant and Weather Forecast from Siri were significantly better than using Voice Action.
- c Note that the relative low number of respondents who reported smooth task completion with Voice Action.

Tasks Completion Google Voice Action	Call a phone #	Navigate to best rated restaurant	Weather Forecast	Set up Alarm @5:00am	Find US Population	Send Email + Dictation
Done smoothly	5	7	6	8	7	9
1st Trial	4	0	2	0	1	0
2nd Trial	3	8	1	2	1	3
3rd Trial	2	0	2	5	2	1
Task Fail	1	0	4	0	4	2

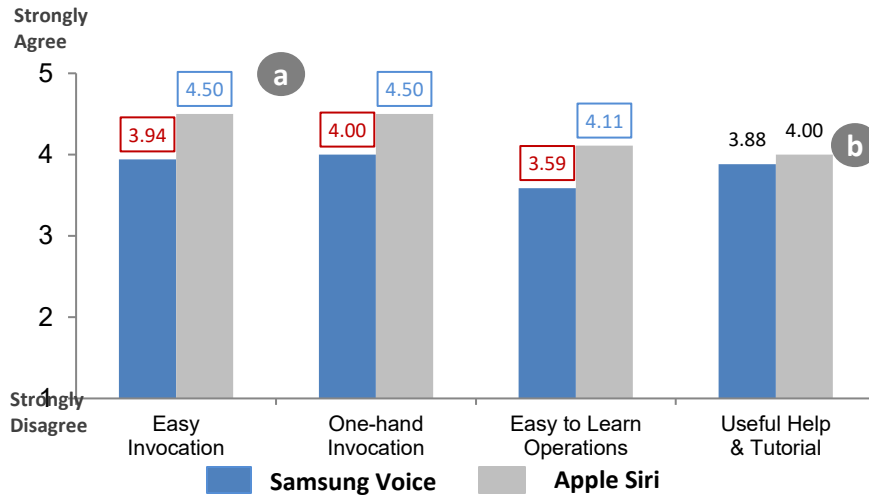
Tasks Completion Apple Siri	Call a phone #	Navigate to best rated restaurant	Weather Forecast	Set up Alarm @5:00am	Find US Population	Send Email + Dictation
Done smoothly	5	12	10	9	8	12
1st Trial	4	0	0	4	1	0
2nd Trial	3	3	5	2	3	1
3rd Trial	2	0	0	0	3	1
Task Fail	1	0	0	0	1	2

Distribution of number of respondents (total N=15)

Initial Usage & Impression: Ease of Invocation and Learning

Initial Impressions Ratings with 5-minute exploration of voice-enabled features

Insight:
Siri invocation follows best design practices. It allows direct and effortless invocation of the speech recognizer. Any design that involves more mental and/or physical effort is likely to defeat the purpose of speech recognition.



a Siri scored significantly higher than Voice talk / command in the ease of invocation, one-hand invocation and ease of learning.

b Siri scored significantly higher than Voice talk / command in the ease of invocation, one-hand invocation and ease of learning.

Initial Product Experience Samsung Voice	Easy Invocation	One-hand Invocation	Easy to Learn Operations	Useful Help & Tutorial
Strongly Agree	5	5	3	5
4	3	9	6	7
3	3	3	6	3
2	3	1	2	2
Strongly Disagree	1	0	0	0

Distribution of number of respondents (total N=18)

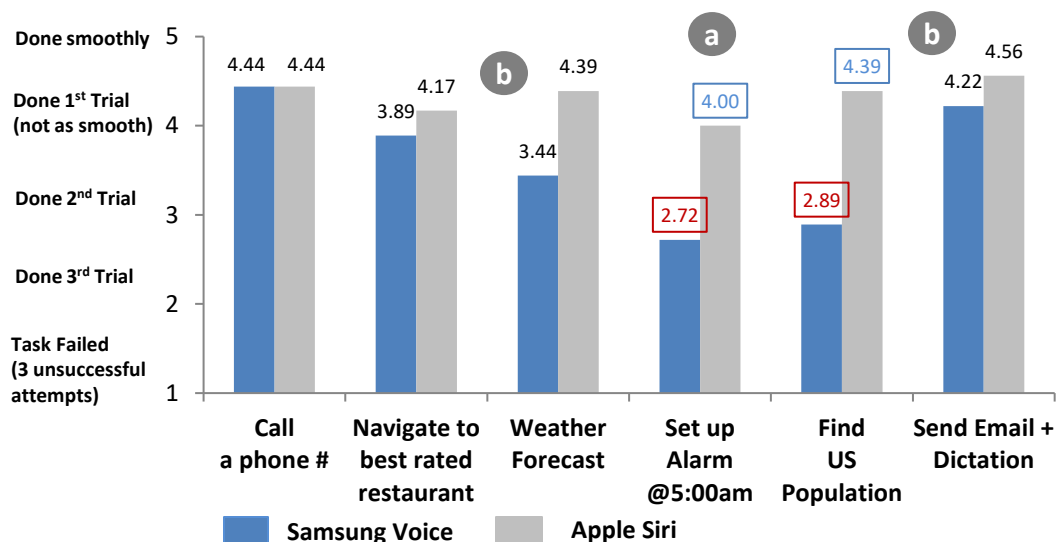
Initial Product Experience Apple Siri	Easy Invocation	One-hand Invocation	Easy to Learn Operations	Useful Help & Tutorial
Strongly Agree	5	11	11	6
4	5	5	8	5
3	2	2	4	5
2	0	0	0	1
Strongly Disagree	1	0	0	0

Task Completion - Overview

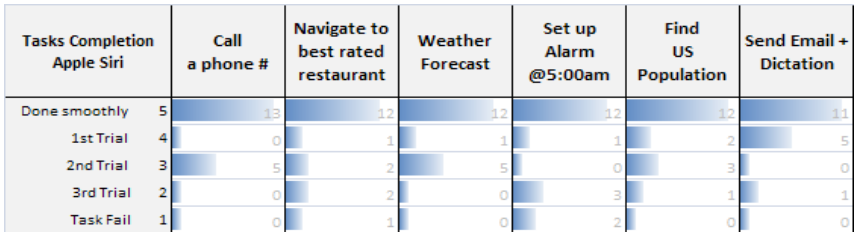
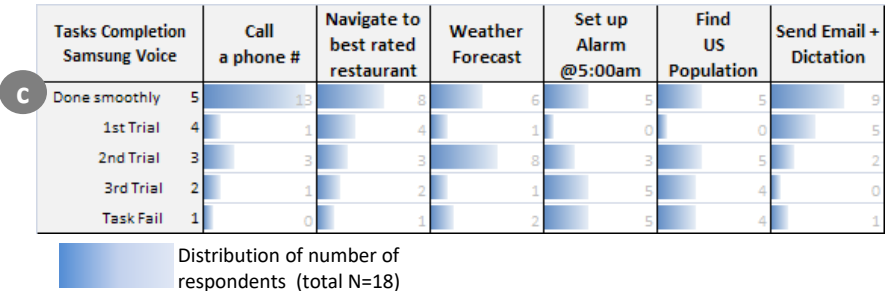
Tasks Completion Rate

Insight:

Consumers can complete more voice-enabled tasks with Siri than Voice Command and Voice talk . This can be attributed to different factors – the flexibility of Siri’s semi-natural language input versus the rigid and limited commands of Samsung Voice as well as functional bugs in Samsung Voice.



- a The completion rates for setting up alarm and finding US population from Siri were significantly better than Voice Command .
- b The completion rates for weather forecast and sending email from Siri were *marginally* significantly better than using Voice Command .
- c Note that the relative low number of respondents who reported smooth task completion with Voice Commands (except calling a phone number).

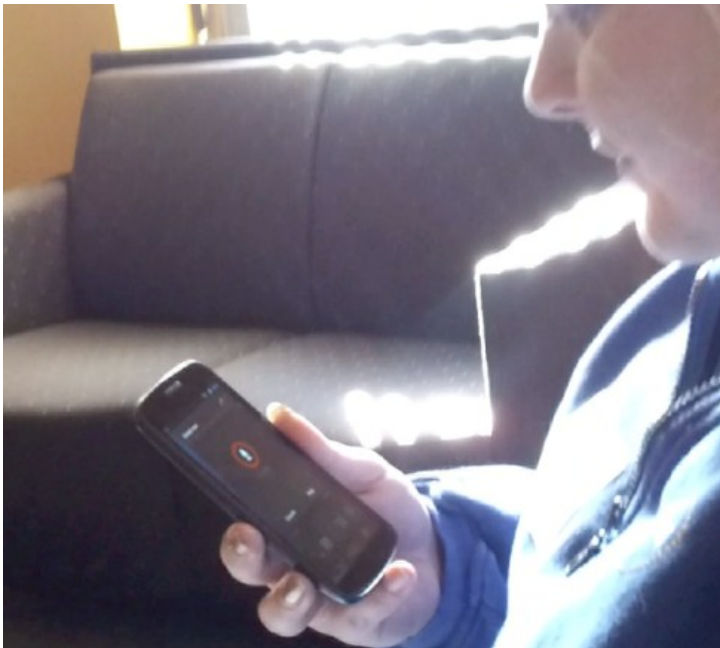


General User Sentiments

Google Voice Action vs. Apple iPhone 4s Siri

Google Voice Action: “the results are not always intuitive there are too many google searches but not enough decision making (e..g, Japanese restaurant search).”

written comment from a respondent dated Nov 7, 2011.



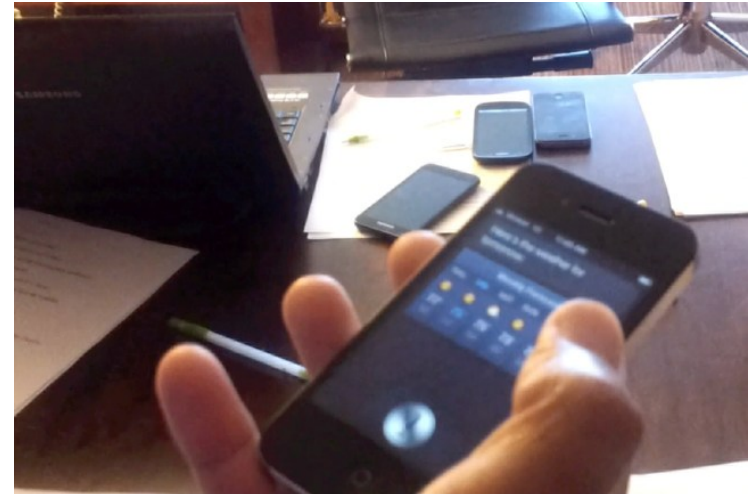
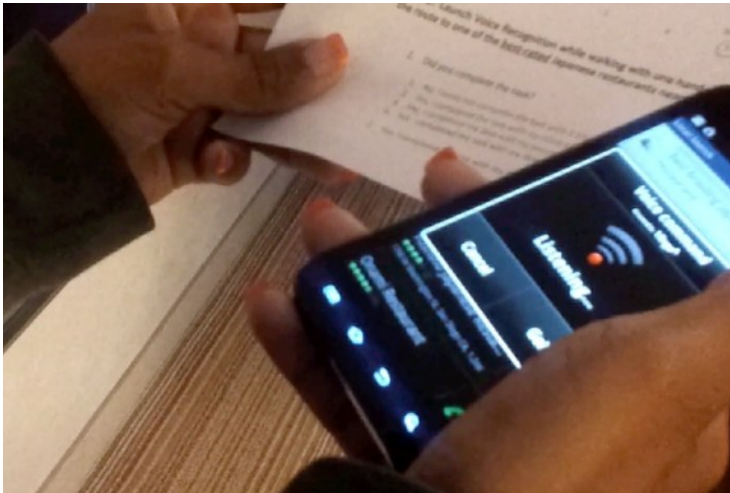
Apple Siri: “Y was so much intuitive, this directing me to exactly where I intended to go; it presented far less errors/inconsistencies and it was much more accurate in interpreting my email via voice recognition.”

written comment from a respondent dated Nov 9, 2011.

General User Sentiments

Galaxy SII Voice Command and Voice Talk vs. Apple iPhone 4s Siri

Samsung Voice: “somewhat inaccurate - rigid command; structure is hard to memorize completely”
written comment from a respondent dated Oct 30, 2011.



Apple Siri: “it is very useful. However, it doesn't recognize all vocabulary, but it is usually able to perform any task if you find the right wording”
written comment from a respondent dated Oct 26, 2011.

Google Voice Action vs. Apple Siri

PRODUCT EXPERIENCE DESIGN REVIEW

Galaxy SII Voice talk and Voice command

Invocation and Speech Recognition Applications

Insight:

Galaxy SII voice features required too much cognitive and perceptual-motor effort/attention resource to activate. It also required too much effort to learn as users had to stick closely to the grammatical rules of the recognizer.

a Default Configuration (Steps to launch Voice App)

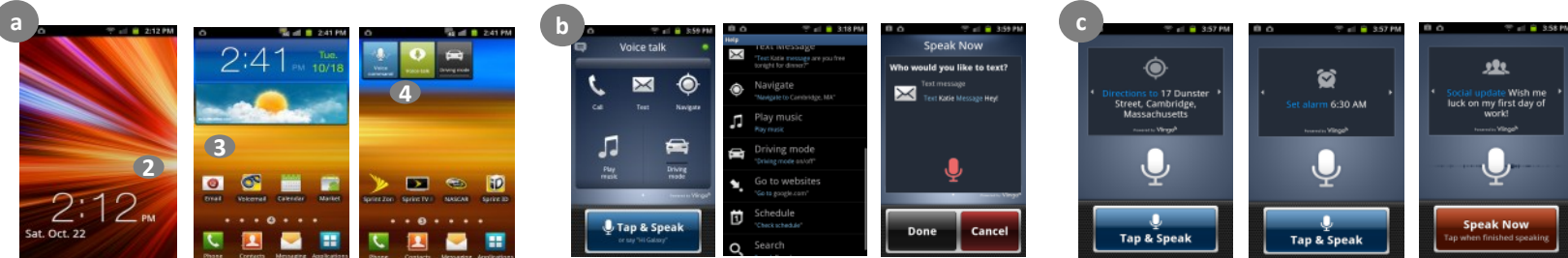
1. Press power/lock button press to wake up device
2. Swipe to unlock screen
3. Swipe to HomeScreen #3 (skipped if user customizes homescreen)
4. Tap Voice talk or Voice command from Voice Widget

b Voice talk

- Voice talk can be used to accomplish 8 different activities with voice only (call, text, navigate, play music, driving mode, go to websites, schedule and search). However, these activities were restricted with the use of voice control. One could not, for example, stop music playing or set up an appointment.
- The 5 handsfree activities supported during driving are highlighted.
- The speech recognizer can be activated by either tapping on the Tap & Speak button or saying "Hi, Galaxy" or by a motion gesture (double tap).

c Voice command

- Voice commands can be used to accomplish more than 20 different activities. They have to be uttered according to a set of restricted syntactical rules.
- 28 examples were built in – one command was visible at a time with horizontal scrolling to review another command. This screen design had misled a number of users to think that they would need to scroll to specific keyword to speak the command.
- Many of the voice-enabled apps required interacting with TouchScreen to complete (e.g., email).
- The speech recognizer can be activated by tapping on Tap & Speak only.
- Most users preferred combining Voice talk and Voice command.

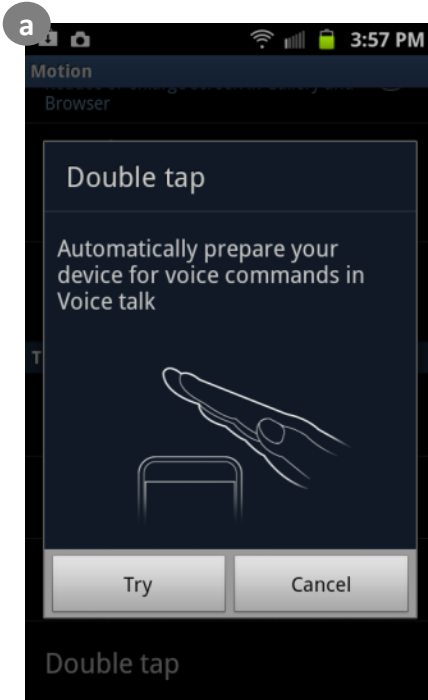


Galaxy SII Voice talk and Voice command

Double tap did not go far enough

Insight:

Motion gesture like “Double tap” is a good example of using gross motion to minimize attention resource to invoke speech recognizer. The primary use of speech recognizer is to complete action with less visual attention. “Double tap” or similar gesture should be extended to context-independent invocation of speech recognizer, not just speech input when the Voice talk app is running.



a

This gesture is useful only when the device is mounted inside a vehicle. Another simple motion gesture should be used to invoke the speech recognizer when the device is not mounted. Users should be able to complete the gesture with one-hand.

Galaxy SII Voice command & Voice talk

Example of Attempts of Contextual Understanding

Insight:

Voice command handled “context” information partially. Machine understanding of contexts is a challenging discipline, requiring long-term vision and investments in both basic and applied R&D .

a

User intention to set alarm at 5:00am tomorrow.
Utterance: “Restaurants” (or similar utterances)

b

User narrowing of choices to Japanese restaurants.
Utterance: “Japanese” (note that absence of “Restaurants”)

c

User wanted best rated Japanese restaurants.
Utterance: “Best rated” (note the absence of “Japanese Restaurants”).
It led to local listing of different stores instead with no apparent connection to best rated or Japanese Restaurants. In this case, one can argue that the best rated Japanese restaurants had already been shown in b.) and this utterance is unnecessary.

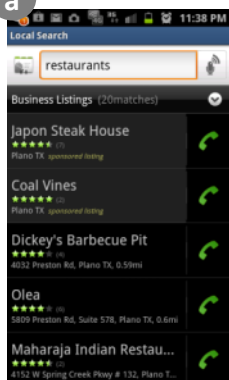
d

Utterance: “Best rated Japanese restaurants”
It led to 2 listing of Insurance agencies then followed by Japanese restaurants (unsorted)

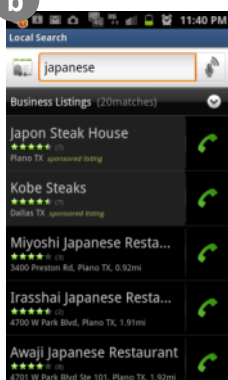
e

Utterance: “Best rated Japanese restaurants” with Siri
It led to 13 Japanese restaurants sorted by rating.

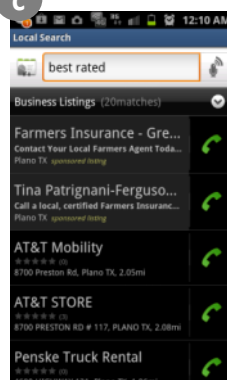
a



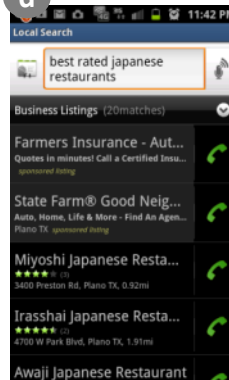
b



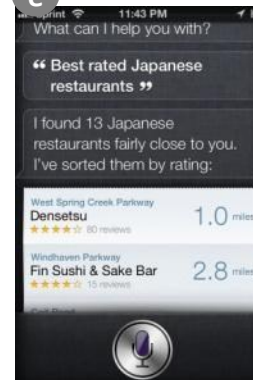
c



d



e



Google Voice Action (Galaxy Neuxs)

Invocation and Speech Recognition Applications

Insight:

Galaxy Nexus voice features required too much cognitive and perceptual-motor effort/attention resource to activate. It also required too much effort to learn as users had to stick closely to the grammatical rules of the recognizer.

a Default Configuration (Steps to launch Voice App)

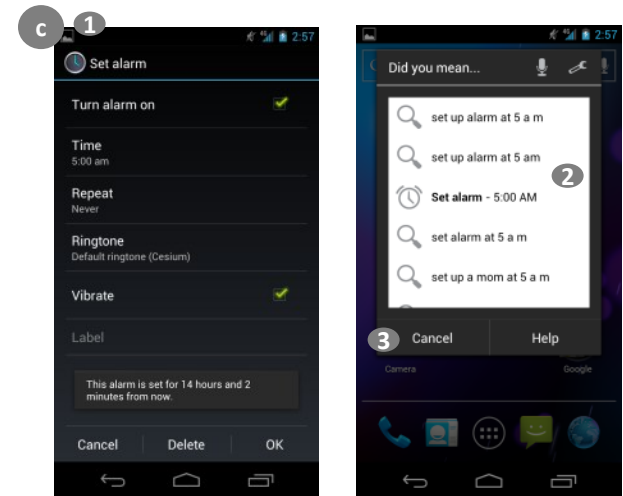
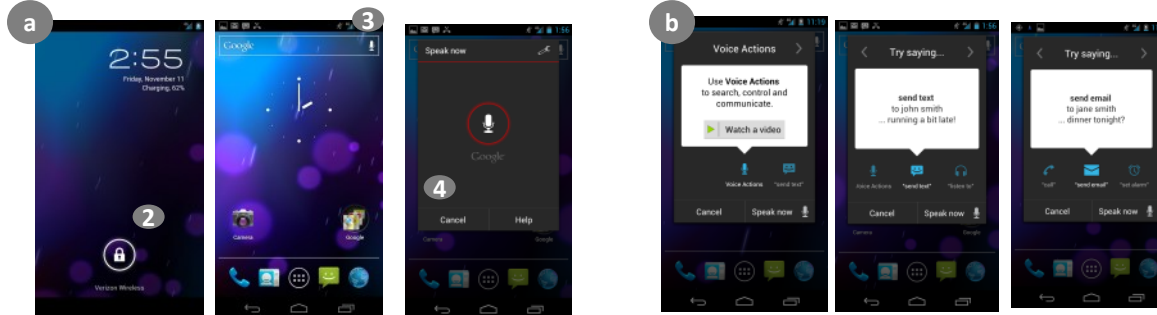
1. Press power/lock button press to wake up device
2. Swipe to unlock screen
3. Press onscreen microphone icon (top right screen) in Google Widget. The screen prompt "Speak Now" appear and user can speak their commands or keywords.

b Voice Action

- Voice Actions include sending text, listening to music, navigation, call contact or phone number or business, send email, set alarm, request map, go to a web site, note to self and web search (12 keywords highlighted in tutorial).
- 12 examples are built in the Help – one command is visible at a time with horizontal scrolling to review another command. They have to be uttered according to a set of restricted syntactical rules.
- Many of the voice-enabled apps may require interacting with TouchScreen to complete (e.g., email).

c Voice Action

1. If the voice input matched a command, the command will be run. The application corresponding to the command will be invoked. The action may be completed automatically or may require additional touch input.
2. "Unrecognized" speech input will be presented in a "Did you mean" Pop-up list of possible candidates (n-best candidates of guessed command or search term) for the utterance.
3. User would either select a candidate or back out to start another command or search.



Google Voice Action (Galaxy Nexus)

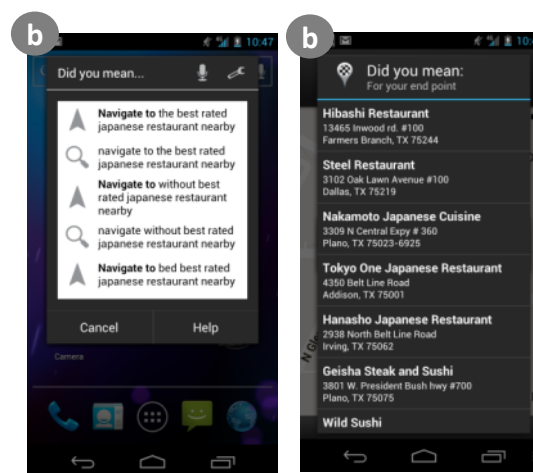
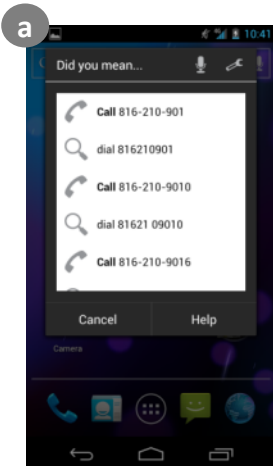
Contextual Guessing with User Confirmation

Insight:

Voice Action used the “Did you mean” pop-up dialog to guess and/or narrow the user intention by providing its n-best guesses of the speech input when the input does not match any command (below the match criteria of Voice Action).

- a
1. User intention to call 816-210-9010
Utterance: “Dial 816-210-9010” (the speech input was not clear enough for Voice Action to carry out the command).
 2. Voice Action displayed the “Did you mean” dialog box with its “n-best contextual” guess candidates of user intention with items of “dial...” presented as possible search candidate and “call...” as possible call voice action items. Users will need to select /confirm their “context” with the item by touch input.

- b
1. Utterance: “Navigate to the best rated Japanese restaurants”
 2. Voice Action displayed the “Did you mean” dialog box with its “n-best contextual” guess candidates of user intention with the same items as possible candidates for Navigation or Google Search.
 3. Selecting Navigation will display another “Did you mean” dialog box with a list of Japanese restaurants (no ratings shown though) for user to select from.



Apple iPhone 4S Siri

Invocation and Speech Recognition Applications

Insight:

Apple iPhone 4S Siri has a single but single step to invoke speech recognizer from virtually anywhere (any app) in the user interface. This design is the pinnacle of minimal but effective design. It minimizes both the attention and effort to invoke speech and makes its user interface transparent and trivial to learn. This also signifies the primacy of natural multimodal interactions in the vision of Apple for the future.

a

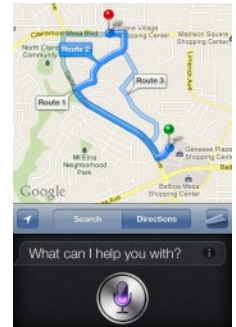
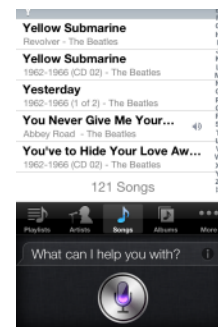
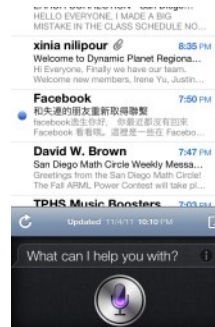
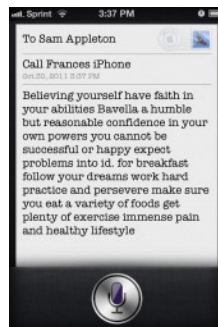
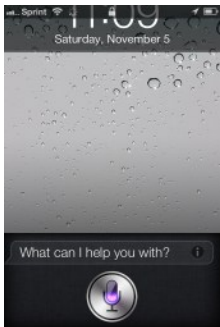
Default Configuration

- When locked: Press and hold [Home] hard button to invoke Siri
- When unlocked: Press and hold [Home] hard button
- In virtually any applications: Press and hold [Home] hard button
- Bring handset to the ear (using proximity sensor)

b

Siri

- Siri can be invoked from any application. This user interface framework can easily enable the use of Siri by any application in context.
- Siri keeps track of the interaction dialog/conversation in the session so that users can scroll to review the entire conversation. This is a key strategic design to facilitate both understanding of users and Siri conversation in the application context (or starting from a new context). It helps both Apple and the users to understand each other.



Apple iPhone 4S Siri Design Highlights

Other Examples

Insight:

Siri has the capability of completing actions on behalf of other applications using multimodal input without switching context.

a

Siri has a set of example scenarios with different examples of utterances that can be used.

b

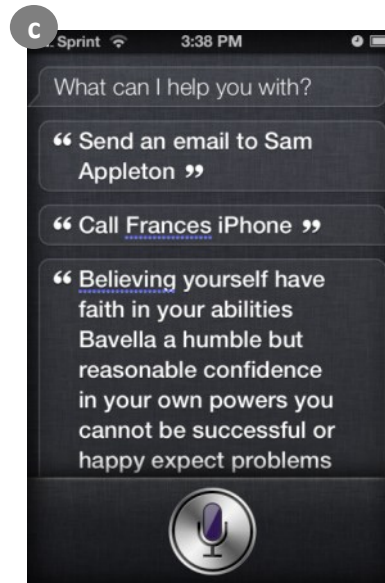
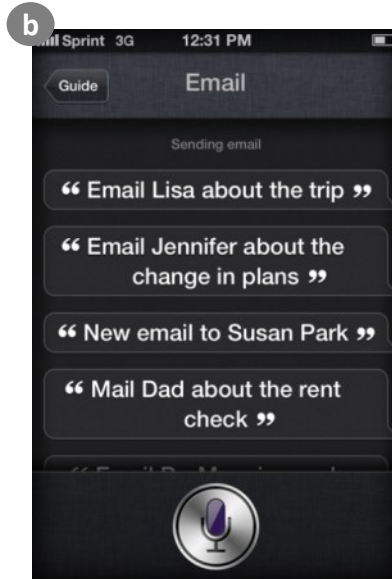
Examples of how an email can be dictated and sent.

c

The text-to-speech app used by Apple is relatively accurate compared to the one in Google Voice Action. It also supports continuous dictation robustly (up to about 30s tested).

d

Siri provided both audio speech confirmation and visual user interface without leaving the session. Respondents can turn off an alarm by touch or confirm by speech or touch.



Apple iPhone 4S Siri Design Highlight

Example of Contextual Understanding

Insight:

Siri understands simple contexts and uses the knowledge effectively in applications like search or reminders.

a

Siri is working with the implicit assumption that user intention is to find local restaurants.

User intention to find Restaurants.

Utterance: "Restaurants" (or similar utterances)

b

User narrowing of choices to Japanese restaurants.

Utterance: "Japanese" (note the absence of "Restaurants")

c

User wanted best rated Japanese restaurants.

Utterance: "Best rated" (note the absence of "Japanese Restaurants")

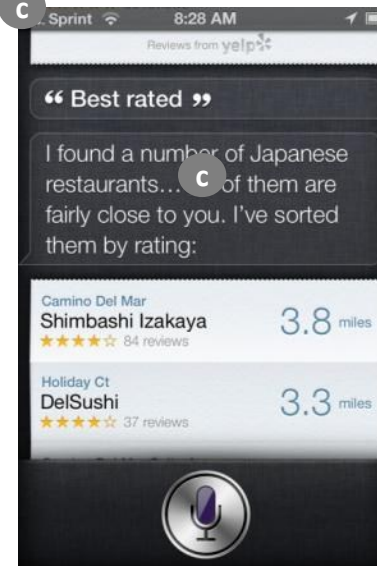
a



b



c



Research Insight from Consumer Behavioral Research

Opportunities: Design Recommendations

Insight:

Direct and effortless invocation of front-end speech recognizer is the first (and easiest) product design direction – a dedicated physical button (and/or gross motion-based gesture) should be included in competitive product.

Physical Design/Ergonomics

1. The front-end speech recognizer can be invoked directly with a single button from both locked screen. A physical button is much more preferred to touch screen invocation. This can be a dedicated side button (physical).
2. A motion-based gesture can be used conveniently to invoke speech recognizer instead of touching the display. The double-tap on Galaxy SII to wake up the speech recognizer is a viable approach.*

User Experience Design & Speech Recognizer Natural/Artificial Intelligence Capability

1. The development /research of natural/artificial intelligence knowledge base and capabilities are critical to compete effectively in contextual understanding. This is also the most difficult and challenging areas. Understanding of user intention and designing products effectively to satisfy user needs require intersection of social/human sciences and engineering.

Application, Conversation & Dialog Design

1. The experience design of Siri dialog, context and multi-modal interactions permeate with modern linguistic theories and conversation principles. Pursuing superficial design without knowledge in psycholinguistic and sociolinguistics will only lead to mediocre product experience.
2. The Voice Action input should be clearly labeled in the Widget and the widget can be renamed as Voice Action/Search. The current Voice Action features on Voice Search 3.0.0 is not discoverable.

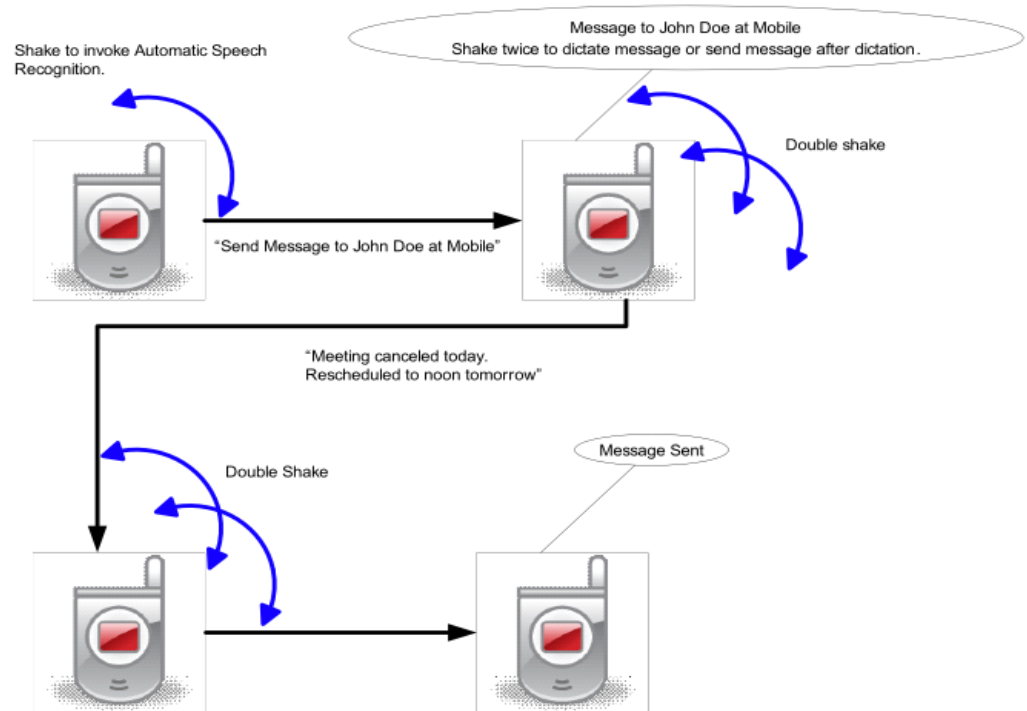
*The double-tap of Galaxy SII is one of the two more useful motion-enabled features.

Physical Ergonomics Design

Design Recommendations

Insight:
Siri can be invoked from anywhere by design while Google Voice Action can only be invoked when the widget is present.

One way that Samsung can compete effectively in ease of speech recognizer invocation with Siri is to use a context-independent gesture (or button) to invoke speech recognizer from anywhere (e.g., the use of a dedicated motion gesture as shown).



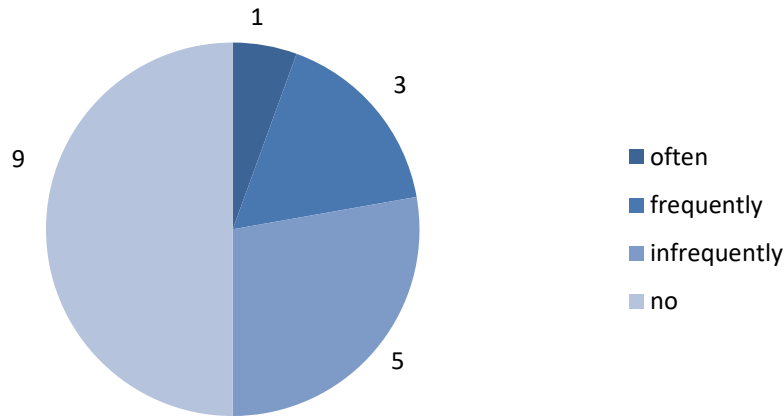
* from STA User Experience Motion/Speech User Interface Design Notes dated Sept 2, 2008

Participants Demographics: Samsung Voice Command/Voice talk vs. Apple Siri

Smartphone owners & intenders

Name	Gender	phone model	Use speech recognition apps mobile?
Justin	M	Motorola Droid	frequently
Rees	M	Motorola Droid	frequently
Josh H.	M	Apple iphone 3GS	infrequently
Lindsay R.	F	Apple iphone4	No
Chris U.	M	Samsung Instinct	No
Marvin H	M	BB Curve/Torch	No
Mark P	M	BB Curve	infrequently
Roberto A	M	Samsung Stunt	No
Carson R.	M	Apple iphone3GS	infrequently
Bobby A.	M	Apple iphone4	No
Ramone L.	M	Samsung Epic	No
Sharelle O.	F	Samsung Epic	No
Laura P.	F	Samsung Epic	frequently
Latoya F.	F	Samsung Vibrant	often
Reginald	M	HTC Thunderbolt	infrequently
Isabella	F	Motorola Flipside	No
Jonathan	M	Motorola Clamshell	infrequently
Frances	F	Motorola Clamshell	No

Use Speech Recognition?



Participants Demographics – Google Voice Action vs. Apple Siri

Smartphone owners & intenders

Name	Gender	phone model	Use speech recognition apps mobile?
Lauren	F	iPhone 4s	often
Scott W.	M	iPhone 4s	infrequently
Keevin L.	M	Samsung Galaxy S (Verizon)	infrequently
Jay A.	M	iPhone 4s	No
Mary S.	F	HTC Incredible	No
Peter W.	M	Motorola W755	infrequently
John W.	M	iPhone 4	No
Mary L.	F	LG UX5600	No
Deflores	F	Samsung Exhibit	No
Stephen P	M	Samsung Continuum	often
Madelyn D.	F	Samsung SGH T-359	No
Daria C	F	HTC Incredible	No
Paul M	M	Motorola Droid Bionic	often
Stephanie	F	Nokia Astound	No
TomP.	M	HTC Hero	infrequently

Use Speech Recognition?

